Git & GitHub Interview Questions

Q1. What is Git?

Answer: Git is a distributed version control system (DVCS) that helps track code changes, collaborate with teams, roll back to older versions, and manage branching/merging.

Pro Tip: Always mention "distributed" – it highlights Git's offline capability.

Q2. What is GitHub?

Answer: GitHub is a cloud platform to host Git repositories and collaborate using Pull Requests, Issues, Actions (CI/CD), and project boards.

Pro Tip: Say Git = tool, GitHub = platform.

Q3. Common Git commands?

Answer:

- git init → Initialize a new Git repository
- git clone <url> → Copy repository from remote to local
- git status → Show modified/untracked files
- git add . → Stage all changes
- git commit -m "msg" → Save a snapshot with a message
- git log --oneline → Show commit history in short form
- git branch → List branches
- git checkout -b <branch> → Create + switch to a new branch
- git merge <branch> → Merge branch into current
- git push origin main → Push local commits to GitHub
- git pull → Fetch + merge changes from remote
- git fetch → Only download changes (no merge)

Pro Tip: In interviews, mention git commit -am "msg" as a shortcut for tracked files.

Q4. Difference between Git and GitHub?

Answer:

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Answer (Q4):

Git is a version control system that runs locally on your machine and helps you track changes in your code.

GitHub is a cloud platform that uses Git. It provides remote repositories, collaboration tools, pull requests, issues, and more.

Pro Tip: Think: Git = tool, GitHub = platform.

Q5. What is a branch in Git?

Answer: A branch is a lightweight movable pointer to a commit. It allows you to work on features or bug fixes independently.

Pro Tip: Use meaningful branch names like "feature/login-page".

Q6. How do you create and switch to a new branch?

Answer:

- git branch <branch-name> → Create a new branch
- git checkout <branch-name> → Switch to branch
- or: git checkout -b <branch-name> → Create & switch

Pro Tip: Newer Git versions also support git switch -c <branch-name>.

Q7. What is merging in Git?

Answer: Merging combines changes from one branch into another.

- git checkout main
- git merge <branch-name>

Pro Tip: Always pull the latest changes before merging to avoid conflicts.

Q8. What is a merge conflict?

Answer: A merge conflict occurs when Git cannot automatically resolve differences in code between two branches.

- Manually resolve conflicts in files
- git add <file>
- git commit

Pro Tip: In interviews, mention git commit -am "msg" as a shortcut for tracked files.

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Q9. What is a pull request?

Answer: A pull request is a way to propose changes from one branch to another and request that those changes be reviewed and merged.

Pro Tip: Write clear PR descriptions and link related issues for better reviews.

Q10. What is stash in Git?

Answer: Stash temporarily saves your uncommitted changes so you can work on something else and reapply them later.

Pro Tip: Use git stash list to see stashed changes and git stash pop to reapply and remove.

Q11. What is the difference between git reset, revert, and restore?

Answer:

- git reset → Move HEAD and branch to another commit. Can rewrite history.
- git revert → Create a new commit that undoes changes. Safe for shared branches.
- git restore → Discard changes in working directory or staging area.

Pro Tip: Use revert for public history, reset for local cleanup.

Q12. What is fork in GitHub?

Answer: Forking creates a copy of a repository under your GitHub account. It's commonly used to contribute to open source projects.

Pro Tip: Keep your fork up to date with the original repository using upstream remote.

Q13. What is the purpose of .gitignore file?

Answer: .gitignore file specifies intentionally untracked files that Git should ignore.

Pro Tip: Add build files, logs, environment files, and IDE settings to .gitignore.

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Q14. What is a git tag?

Answer: A tag is used to mark a specific point in history as important, usually for release versions.

Pro Tip: Use annotated tags with messages for better clarity.

Q15. What is a remote in Git?

Answer: A remote is a version of your repository that is hosted on the internet or network (e.g., GitHub).

Pro Tip: Use meaningful remote names like "origin".

Q16. What is the difference between git pull and git fetch?

Answer:

- git fetch → Downloads changes from remote but doesn't merge.
- git pull → Fetches changes and merges into your current branch.

Pro Tip: Use fetch to review changes before merging.

Q17. How do you resolve merge conflicts?

Answer:

1. Git will mark conflicting files.
2. Open the file and look for conflict markers (<<<<<<<, ======, >>>>>>).
3. Edit the file to keep the correct changes.
4. Save the file and run git add <file>.
5. Commit the merge.

Pro Tip: Communicate with your team to avoid conflicts.

Q18. What is GitHub Actions?

Answer: GitHub Actions is a CI/CD tool that allows you to automate workflows like testing, building, and deploying your code.

Pro Tip: Store workflow files in the .github/workflows directory.